Mathematical Analysis II
Week 9 Homework (April 30)

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Section 10.3

15. 求由曲面 $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=\dfrac{z^2}{c^2}$ 和 $z=c$ 所围成的均匀物体的重心坐标.

解. 由于区域关于 z 轴旋转对称且密度均匀, 因此仅需计算 z 坐标 $\dfrac{\iiint_V z\,\mathrm{d}V}{\iiint_V \mathrm{d}V}$.

采用广义柱面坐标变换: 记

$$\begin{cases} x = ar\cos\theta \\ y = br\sin\theta \\ z = z \end{cases},$$

则积分区域为 $\begin{cases} 0\le r\le \dfrac{z}{c} \\ 0\le\theta\le 2\pi \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,z)}=abr$.

因此

$$\begin{aligned}\iiint_V z\,\mathrm{d}V &= ab\int_0^c z\,\mathrm{d}z\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{z}{c}} r\,\mathrm{d}r \\ &= \frac{ab\pi}{c^2}\int_0^c z^3\,\mathrm{d}z \\ &= \frac{abc^2\pi}{4}\end{aligned}$$

$$\begin{aligned}\iiint_V \mathrm{d}V &= ab\int_0^c \mathrm{d}z\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{z}{c}} r\,\mathrm{d}r \\ &= \frac{ab\pi}{c^2}\int_0^c z^2\,\mathrm{d}z \\ &= \frac{abc\pi}{3}\end{aligned}$$

从而 z 坐标为 $\frac{\frac{abc^2\pi}{4}}{\frac{abc\pi}{3}}=\frac{3c}{4}$.

因此重心坐标为 $\left(0,0,\frac{3c}{4}\right)$.

16. 设球体 $x^2+y^2+z^2\leq 2az$ 内各点密度与各点到原点的距离成反比, 求其重心坐标.

解. 原方程可化为 $x^2+y^2+(z-a)^2\leq a^2$, 表示球心为 $(0,0,a)$、半径为 a 的球体. 由于区域关于 z 轴旋转对称, 因此仅需计算 z 轴坐标. 设密度与到原点距离的比例系数为 k, 即点 (x,y,z) 处的密度为 $\frac{k}{\sqrt{x^2+y^2+z^2}}$.

采用球面坐标变换: 记

$$\begin{cases} x=r\sin\theta\cos\varphi \\ y=r\sin\theta\sin\varphi \\ z=r\cos\theta \end{cases},$$

则积分区域为 $\begin{cases} 0\leq r\leq 2a\cos\theta \\ 0\leq\theta\leq\frac{\pi}{2} \\ 0\leq\varphi\leq 2\pi \end{cases}$, 雅可比行列式为 $\frac{\partial(x,y,z)}{\partial(r,\theta,\varphi)}=r^2\sin\theta$.

因此

$$\begin{aligned}\iiint_V z\rho\left(\overrightarrow{r}\right)\,\mathrm{d}V &= k\int_0^{\frac{\pi}{2}}\sin\theta\cos\theta\,\mathrm{d}\theta\int_0^{2\pi}\,\mathrm{d}\varphi\int_0^{2a\cos\theta}r^2\,\mathrm{d}r \\ &= \frac{16a^3k\pi}{3}\int_0^{\frac{\pi}{2}}\sin\theta\cos^4\theta\,\mathrm{d}\theta \\ &= \frac{16a^3k\pi}{15}\end{aligned}$$

$$\begin{aligned}\iiint_V \rho\left(\overrightarrow{r}\right)\,\mathrm{d}V &= k\int_0^{\frac{\pi}{2}}\sin\theta\,\mathrm{d}\theta\int_0^{2\pi}\,\mathrm{d}\varphi\int_0^{2a\cos\theta}r\,\mathrm{d}r \\ &= 4a^2k\pi\int_0^{\frac{\pi}{2}}\sin\theta\cos^2\theta\,\mathrm{d}\theta \\ &= \frac{4a^2k\pi}{3}\end{aligned}$$

从而 z 轴坐标为 $\frac{\frac{16a^3k\pi}{15}}{\frac{4a^2k\pi}{3}}=\frac{4a}{5}$.

因此重心坐标为 $\left(0,0,\frac{4a}{5}\right)$.

18. 求以下各物体的转动惯量 (设 $\rho=$ 常数)

(1) 质量为 m, 半径为 R 的薄圆盘, 对于:

(a) 通过圆心并垂直圆盘的轴;

解. 取 z 轴为转轴.

需计算 $\iint_D d^2\rho\,\mathrm{d}A$.

采用极坐标变换: 记

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases},$$

则积分区域为 $\begin{cases} 0 \le r \le R \\ 0 \le \theta \le 2\pi \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y)}{\partial(r,\theta)} = r$, 且 $d^2 = r^2$.

面密度 $\rho = \dfrac{m}{\pi R^2}$.

因此

$$\begin{aligned} \iint_D d^2\rho\,\mathrm{d}A &= \frac{m}{\pi R^2}\int_0^{2\pi}\,\mathrm{d}\theta\int_0^R r^3\,\mathrm{d}r \\ &= \frac{mR^2}{2} \end{aligned}$$

(b) 直径.

解. 取 x 轴为转轴.

此时 $d^2 = y^2 = r^2\sin^2\theta$.

因此

$$\begin{aligned} \iint_D d^2\rho\,\mathrm{d}A &= \frac{m}{\pi R^2}\int_0^{2\pi}\sin^2\theta\,\mathrm{d}\theta\int_0^R r^3\,\mathrm{d}r \\ &= \frac{mR^2}{4} \end{aligned}$$

(2) 质量为 m, 半径为 R 的球体, 对于通过球心的轴线.

解. 取 z 轴为转轴, 需计算 $\iiint_V d^2\rho\,\mathrm{d}V$.

采用球面坐标变换: 记

$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases},$$

则积分区域为 $\begin{cases} 0 \le r \le R \\ 0 \le \theta \le \pi \\ 0 \le \varphi \le 2\pi \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,\varphi)} = r^2\sin\theta$, 且 $d^2 = r^2\sin^2\theta$.

体密度 $\rho = \dfrac{m}{\frac{4}{3}\pi R^3} = \dfrac{3m}{4\pi R^3}$.

因此

$$
\begin{aligned}
\iiint_V d^2\rho \,\mathrm{d}V &= \frac{3m}{4\pi R^3}\int_0^{2\pi} \mathrm{d}\varphi \int_0^{\pi} \sin^3\theta \,\mathrm{d}\theta \int_0^R r^4 \,\mathrm{d}r \\
&= \frac{3m}{4\pi R^3}\cdot 2\pi \cdot \frac{4}{3}\cdot \frac{R^5}{5} \\
&= \frac{2}{5} mR^2
\end{aligned}
$$

19. 求密度为 ρ 的均匀球锥体对于在其顶点为一单位质量的质点的吸引力, 设球的半径为 R, 而轴截面的扇形的角等于 2α.

解. 取 z 轴为对称轴, 顶点为原点. 由于垂直方向上的引力分量相互抵消, 因此仅需计算沿 z 轴方向的引力分量.

采用球面坐标变换: 记

$$
\begin{cases}
x = r\sin\theta\cos\varphi \\
y = r\sin\theta\sin\varphi \\
z = r\cos\theta
\end{cases},
$$

则积分区域为 $\begin{cases} 0 \le r \le R \\ 0 \le \theta \le \alpha \\ 0 \le \varphi \le 2\pi \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,\varphi)} = r^2\sin\theta$. 位置 (r,θ,φ) 处质量元对质点的引力在 z 轴方向的分量为 $\dfrac{G\rho}{r^2}\cos\theta$.

因此

$$
\begin{aligned}
\iiint_V \frac{G\rho}{r^2}\cos\theta \,\mathrm{d}V &= G\rho \int_0^{2\pi} \mathrm{d}\varphi \int_0^{\alpha} \sin\theta\cos\theta \,\mathrm{d}\theta \int_0^R \mathrm{d}r \\
&= \pi R G\rho \sin^2\alpha
\end{aligned}
$$

Section 10.4

1. 计算下列 n 重积分.

(1) $\displaystyle\int\cdots\int_{[0,1]^n} (x_1^2 + \cdots + x_n^2)\,\mathrm{d}x_1\cdots\mathrm{d}x_n$;

解. 首先计算 $\int\cdots\int_{[0,1]^n} x_i^2 \,\mathrm{d}x_1\cdots\mathrm{d}x_n$ $(i = 1, 2, \ldots, n)$:

$$
\begin{aligned}
\int\cdots\int_{[0,1]^n} x_i^2 \,\mathrm{d}x_1\cdots\mathrm{d}x_n &= \int_0^1 \mathrm{d}x_1 \cdots \int_0^1 x_i^2 \,\mathrm{d}x_i \cdots \int_0^1 \mathrm{d}x_n \\
&= 1\cdot\frac{1}{3}\cdot 1 \\
&= \frac{1}{3}
\end{aligned}
$$

可见, 各部分单独积分结果与 i 无关.

因此 $\int\cdots\int_{[0,1]^n}(x_1^2+\cdots+x_n^2)\,\mathrm{d}x_1\cdots\mathrm{d}x_n=\dfrac{n}{3}$.

(2) $\displaystyle\int\cdots\int_{[0,1]^n}(x_1+\cdots+x_n)^2\,\mathrm{d}x_1\cdots\mathrm{d}x_n$;

解. 展开得 $(x_1+\cdots+x_n)^2=\sum x_i^2+2\sum_{i\neq j}x_ix_j$.

对于 n 个 x_i^2 项, 由 (1) 可得 $\int\cdots\int_{[0,1]^n}(x_1^2+\cdots+x_n^2)\,\mathrm{d}x_1\cdots\mathrm{d}x_n=\dfrac{n}{3}$.

对于 $\dfrac{n(n-1)}{2}$ 个 x_ix_j 项 $(i\neq j)$, 有

$$\begin{aligned}\int_0^1\mathrm{d}x_i\cdots\int_0^1\mathrm{d}x_i\cdots\int_0^1 x_ix_j\,\mathrm{d}x_j\cdots\int_0^1\mathrm{d}x_n&=\int_0^1\mathrm{d}x_i\int_0^1 x_ix_j\,\mathrm{d}x_j\\&=\frac{1}{2}\int_0^1 x_i\,\mathrm{d}x_i\\&=\frac{1}{4}\end{aligned}$$

因此 $\int\cdots\int_{[0,1]^n}2\sum_{i\neq j}x_ix_j\,\mathrm{d}x_1\cdots\mathrm{d}x_n=\dfrac{n(n-1)}{4}$.

从而总和为 $\dfrac{n}{3}+\dfrac{n(n-1)}{4}=\dfrac{n(3n+1)}{12}$.

(3) $\displaystyle\int_0^1\mathrm{d}x_1\int_0^{x_1}\mathrm{d}x_2\cdots\int_0^{x_{n-1}}x_1\cdots x_n\,\mathrm{d}x_n$.

解.

$$\begin{aligned}\int_0^1\mathrm{d}x_1\int_0^{x_1}\mathrm{d}x_2\cdots\int_0^{x_{n-1}}x_1\cdots x_n\,\mathrm{d}x_n&=\int_0^1 x_1\,\mathrm{d}x_1\int_0^{x_1}x_2\,\mathrm{d}x_2\cdots\int_0^{x_{n-1}}x_n\,\mathrm{d}x_n\\&=\int_0^1 x_1\,\mathrm{d}x_1\int_0^{x_1}x_2\,\mathrm{d}x_2\cdots\int_0^{x_{n-2}}\frac{1}{2}x_{n-1}^3\,\mathrm{d}x_{n-1}\\&=\int_0^1 x_1\,\mathrm{d}x_1\int_0^{x_1}x_2\,\mathrm{d}x_2\cdots\int_0^{x_{n-3}}\frac{1}{2\times4}x_{n-2}^5\,\mathrm{d}x_{n-2}\\&=\dots\\&=\int_0^1\frac{1}{2^{n-1}(n-1)!}x_1^{2n-1}\,\mathrm{d}x_1\\&=\frac{1}{2^n n!}\end{aligned}$$

3. 设 $f(x)$ 连续, 证明:

$$\int_0^a\mathrm{d}x_1\int_0^{x_1}\mathrm{d}x_2\cdots\int_0^{x_{n-1}}f(x_n)\,\mathrm{d}x_n=\frac{1}{(n-1)!}\int_0^a f(t)(a-t)^{n-1}\,\mathrm{d}t.$$

证. 积分区域为 $0\le x_n\le x_{n-1}\le\cdots\le x_1\le a$. 由于最内层被积函数仅与 x_n 有关, 固定 $x_n=t$, 则 $t\le x_{n-1}\le\cdots\le x_1\le a$. 变换积分顺序, 可得

$$\begin{aligned}\int_0^a \mathrm{d}x_1 \int_0^{x_1} \mathrm{d}x_2 \cdots \int_0^{x_{n-1}} f(x_n)\,\mathrm{d}x_n &= \int_0^a f(t)\,\mathrm{d}t \int_{t\le x_{n-1}\le\cdots\le x_1\le a} \mathrm{d}x_1\cdots\mathrm{d}x_{n-1} \\ &= \int_0^a f(t)\frac{(a-t)^{n-1}}{(n-1)!}\,\mathrm{d}t \\ &= \frac{1}{(n-1)!}\int_0^a f(t)(a-t)^{n-1}\,\mathrm{d}t\end{aligned}$$

Q.E.D.

4. 设 $f(x)$ 连续, 证明:

$$\int_0^a \mathrm{d}x_1 \int_0^{x_1} \mathrm{d}x_2 \cdots \int_0^{x_{n-1}} f(x_1)f(x_2)\cdots f(x_n)\,\mathrm{d}x_n = \frac{1}{n!}\left(\int_0^a f(t)\,\mathrm{d}t\right)^n.$$

证.

$$\begin{aligned}&\int_0^a \mathrm{d}x_1 \int_0^{x_1} \mathrm{d}x_2 \cdots \int_0^{x_{n-1}} f(x_1)f(x_2)\cdots f(x_n)\,\mathrm{d}x_n \\ =&\int_{0\le x_n\le\cdots\le x_1\le a} f(x_1)f(x_2)\cdots f(x_n)\,\mathrm{d}x_1\cdots\mathrm{d}x_n \\ =&\frac{1}{n!}\left(\int_0^a f(t)\,\mathrm{d}t\right)^n\end{aligned}$$

Q.E.D.

第 10 章综合习题

8. 设 a, b 是不全为 0 的常数. 求证:

$$\iint\limits_{x^2+y^2\le 1} f(ax+by+c)\,\mathrm{d}x\mathrm{d}y = 2\int_{-1}^1 \sqrt{1-t^2}\, f\left(t\sqrt{a^2+b^2}+c\right)\,\mathrm{d}t.$$

证. 记 $r=\sqrt{a^2+b^2}$, 采用正交变换:

$$\begin{cases} u = \dfrac{ax+by}{r} \\ v = \dfrac{-bx+ay}{r} \end{cases},$$

则 $u^2+v^2\le 1$, 且雅可比行列式 $\dfrac{\partial(x,y)}{\partial(u,v)} = 1$.

此时被积函数化为 $f(ur+c)$.

因此

$$
\begin{aligned}
\iint_{x^2+y^2\le 1} f(ax+by+c)\,\mathrm{d}x\mathrm{d}y &= \int_{u^2+v^2\le 1} f(ur+c)\,\mathrm{d}u\mathrm{d}v \\
&= \int_{-1}^{1}\mathrm{d}u\int_{-\sqrt{1-u^2}}^{\sqrt{1-u^2}} f(ur+c)\,\mathrm{d}v \\
&= 2\int_{-1}^{1}\sqrt{1-u^2}f(ur+c)\,\mathrm{d}u
\end{aligned}
$$

令 $t=u$, 则有

$$
\begin{aligned}
\iint_{x^2+y^2\le 1} f(ax+by+c)\,\mathrm{d}x\mathrm{d}y &= 2\int_{-1}^{1}\sqrt{1-u^2}f(ur+c)\,\mathrm{d}u \\
&= 2\int_{-1}^{1}\sqrt{1-t^2}\,f\left(t\sqrt{a^2+b^2}+c\right)\,\mathrm{d}t
\end{aligned}
$$

Q.E.D.